ETH-tight parameter dependence for Steiner Forest by vertex cover

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Editorial note. Only arxiv:2402.09835:thm:vc counts. All author mathematical text and source-local proofs retained. BibTeX/natbib bibliography rendered from original main.bbl metadata; native unchanged. Unused appendix package and pdftex driver adapted to existing XeLaTeX.

Allowed standard prerequisites (editorial).

- Standard graph algorithms and ETH cited by authors. All source-local proofs and necessary context retained.

Abstract

In this paper we reassess the parameterized complexity and approximability of the well-studied Steiner Forest problem in several graph classes of bounded width. The problem takes an edge-weighted graph and pairs of vertices as input, and the aim is to find a minimum cost subgraph in which each given vertex pair lies in the same connected component. It is known that this problem is APX-hard in general, and NP-hard on graphs of treewidth 3, treedepth 4, and feedback vertex set size 2. However, Bateni, Hajiaghayi and Marx [JACM, 2011] gave an approximation scheme with a runtime of $n^{O(k^2/\varepsilon)}$ on graphs of treewidth k. Our main result is a much faster *efficient parameterized approximation scheme (EPAS)* with a runtime of $2^{O(\frac{k^2}{\varepsilon}\log\frac{k}{\varepsilon})} \cdot n^{O(1)}$. If k instead is the vertex cover number of the input graph, we show how to compute the optimum solution in $2^{O(k \log k)} \cdot n^{O(1)}$ time, and we also prove that this runtime dependence on k is asymptotically best possible, under ETH. Furthermore, if k is the size of a feedback edge set, then we obtain a faster $2^{O(k)} \cdot n^{O(1)}$ time algorithm, which again cannot be improved under ETH.

1 Introduction

The Steiner Forest problem is one of the most well-studied problems in network design [16, 23, 24, 30]. In this problem the input consists of a graph $G = (V, E)$ with positive edge weights, a set of *terminals* $R \subseteq V$, and a set of *demands* $D \subseteq \binom{R}{2}$. The objective is to select a subgraph $F \subseteq G$, minimizing the total cost of selected edges, while ensuring that for every demand pair $\{s, t\} \in D$, s and t are in the same connected component of F. Since edge weights are positive, it is easy to see that the optimal solution is always a forest. The Steiner Forest problem finds many applications (see surveys [11, 30, 34, 35]), for example in telecommunication networks (cf. [35]).

Our goal in this paper is to reassess the complexity of this fundamental problem from the point of view of parameterized complexity and approximation algorithms.[1] In order to recall the context, it is helpful to compare Steiner Forest to the even more well-studied Steiner Tree problem, which is the special case of Steiner Forest where all terminals are required to be connected, i.e., $D = \binom{R}{2}$, and an optimal solution is a tree. Steiner Tree was already included in Karp's seminal list [26] of NP-hard problems from the 1970s. From the approximation point of view, Steiner Tree (and therefore Steiner Forest) is known to be APX-hard [13], but both problems admit constant factor approximations in polynomial time for general input graphs, where the best approximation factors known are $\ln(4) + \varepsilon < 1.39$ [10] and 2 [1, 33], respectively. Despite this similarity, when considering graph width parameters the problems exhibit wildly divergent behaviors from the parameterized complexity point of view: whereas Steiner

[1] We assume the reader is familiar with the basics of parameterized complexity and approximation algorithms, such as the classes FPT and APX and the definition of treewidth, as given in standard textbooks [14, 19, 36]. We give full definitions of all parameters in Section 2.

Tree is FPT parameterized by standard structural parameters such as treewidth and can in fact even be solved in single exponential $2^{O(k)}n^{O(1)}$ time [7] when k is the treewidth, Steiner Forest is NP-hard on graphs of treewidth 3, as shown independently by Gassner [21] and Bateni et al. [4].

Steiner Forest is therefore a problem that presents a dramatic jump in complexity in this context, compared to Steiner Tree, as the hardness result on graphs of treewidth 3 rules out even an XP algorithm for parameter treewidth. One of the main positive contributions of Bateni et al. [4] was an algorithm attempting to bridge this gap using approximation. In particular, they showed that Steiner Forest admits an approximation scheme for graphs of treewidth k, which computes a $(1+\varepsilon)$-approximation in $n^{O(k^2/\varepsilon)}$ time for any $\varepsilon > 0$. Hence, if we allow slightly sub-optimal solutions, we can at least place the problem in XP parameterized by treewidth. In their paper, Bateni et al. [4] remark that because the exponent of the polynomial of this runtime depends on k and ε, "it remains an interesting question for future research whether this dependence can be removed", that is, whether a $(1+\varepsilon)$-approximation can be obtained in FPT time.

The main result of our paper is a positive resolution of the question of Bateni et al. [4]: we show that Steiner Forest admits an *efficient parameterized approximation scheme (EPAS)* for treewidth, that is, a $(1+\varepsilon)$-approximation algorithm with a runtime of the form $f(k,\varepsilon)n^{O(1)}$. In other words, we show that their algorithm can be improved in a way that makes the running time FPT not only in the treewidth, but also in $1/\varepsilon$. More precisely, we show the following:

Theorem 1. *The Steiner Forest problem admits an EPAS parameterized by the treewidth k with a runtime of $2^{O(\frac{k^2}{\varepsilon}\log\frac{k}{\varepsilon})} \cdot n^{O(1)}$.*

Moving on from treewidth, we ask what the most general parameter is for which we may hope to obtain an FPT *exact* algorithm for Steiner Forest. We observe that the NP-hardness result of Bateni et al. [4], Gassner [21] for Steiner Forest on graphs of treewidth 3 actually has some further implications for some even more restricted parameters: the graphs constructed in their reductions also have constant *treedepth* and *feedback vertex set* size, implying that the problem remains hard for both of these parameters (which are incomparable in general). More precisely, known reductions imply the following:

Theorem 2 (Bateni et al. [4], Gassner [21])**.** *The Steiner Forest problem is NP-hard on graphs of treewidth 3, treedepth 4, and feedback vertex set of size 2.*

This leads us to consider even more restricted parameters, such as the size of a *vertex cover* and *feedback edge set*, which are not bounded in this reduction. Indeed, not only do we prove that Steiner Forest is FPT for both of these parameters, but we are also able to determine the correct parameter dependence, under the Exponential Time Hypothesis (ETH). For feedback edge set the optimal dependence is single exponential:

Theorem 3. *The Steiner Forest problem is FPT parameterized by the size k of a feedback edge set and can be solved in $2^{O(k)}n^{O(1)}$ time. Furthermore, no $2^{o(k)}n^{O(1)}$ time algorithm exists, under ETH.*

For the parameterization by the vertex cover size, we obtain a slower runtime for our FPT algorithm. Interestingly, we are also able to prove that this is best possible, under ETH. Our lower bound for Steiner Forest is in contrast to the Steiner Tree problem, for which a faster $2^{O(k)}n^{O(1)}$ time algorithm exists, even if k is the treewidth [7].

Theorem 4. *The Steiner Forest problem is FPT parameterized by the size k of a vertex cover and can be solved in $2^{O(k\log k)}n^{O(1)}$ time. Furthermore, no $2^{o(k\log k)}n^{O(1)}$ time algorithm exists, under ETH.*

We remark that Bodlaender et al. [8] recently independently showed that Steiner Forest admits a $2^{O(k\log k)}n^{O(1)}$ time algorithm for the size k of a vertex cover (improving an algorithm for the unweighted version of the problem given in [22]). While they develop their own dynamic program to solve this problem, we rely on an existing algorithm by [4] (see Theorem 5). Accordingly our description of the algorithm is very short compared to [8]. The more interesting part of Theorem 4 however is the proof of the lower bound.

1.1 Overview of Techniques

Let us briefly sketch the high level ideas of our results given by Theorems 1, 3 and 4.

EPAS for treewidth. Our algorithm extends the work of [4], so let us briefly recall some key ideas. Given a rooted tree decomposition, a terminal t is called *active* for a bag B if there is a demand $\{s,t\} \in D$ such that t lies in the sub-tree rooted at B while s does not (see Section 2 for formal definitions). It is a standard property of tree decompositions that every bag is a separator. Hence the component of any feasible solution that contains an active terminal must intersect B. The hardness of the problem now inherently stems from the fact that we have to decide for all active terminals of a bag, how the corresponding component intersects the bag, and therefore how the active terminals (whose number is unbounded by k) are partitioned into connected components. Suppose, however, that someone supplied us with this information, that is, suppose that for each bag B we are given a set of partitions Π_B of its active terminals and we are promised that the optimal solution *conforms* to all Π_B. By this we mean that if we look at how the optimal solution partitions the active terminals of B into connected components and call this partition π, then $\pi \in \Pi_B$, that is, the optimal partition is always one of the supplied options. In this case, using this extra information, the problem does become tractable, as shown in [4]:

Theorem 5 (Bateni et al. [4])**.** *For an input graph G on n vertices, let a rooted nice tree decomposition of width k be given, such that all terminals lie in bags of leaf nodes of the decomposition. Also, let a set Π_B of partitions of the active terminals of each bag B of the decomposition be given. If $p = \sum_B |\Pi_B|$ is the total number of partitions, then a minimum cost* Steiner Forest *solution conforming to all Π_B can be computed in $2^{O(k \log k)} \cdot (pn)^{O(1)}$ time.*

The above theorem does not seem immediately helpful, since one would still need to find a small collection of partitions Π_B in order to obtain an efficient algorithm. Note however, that the partition sets may conform to an approximate solution as well, which would let the algorithm compute a solution that it at least as good. The strategy of [4] therefore is to construct a collection of partitions that has size polynomial in n (when k, ε are fixed constants) by stipulating that when two active terminals are "close" to each other, they should belong in the same set of the partition of some near-optimal solution. In order to bound the resulting approximation ratio, they need to provide a charging scheme: starting from an optimal solution, they merge components which are "close", to obtain a solution that conforms to the Π_B used by the algorithm. They then show that the resulting solution is still near-optimal by charging the extra cost incurred by a merging operation to one of the two merged components.

A blocking point in the above is that we need to make sure we do not "overcharge" any component. This is accomplished in [4] via a partial ordering of the components: we order the components according to the highest bag of the rooted tree decomposition they intersect, and whenever two components are merged we charge this to the *lower* component. As shown in [4], this ensures that no component is charged for more than k merges. Unfortunately, this also implies that the merging procedure is not symmetric, which severely diminishes the contexts in which we can apply it.

Let us now describe how our approach improves upon this algorithm. A key ingredient will be a more sophisticated charging scheme, which will allow us to obtain a better (smaller) collection of partitions Π_B, without sacrificing solution quality. Counter-intuitively, we will achieve this by introducing a *second* parameter: the height h of the tree decomposition. Informally, we will now construct a near-optimal solution by merging two components whenever the connection cost is low compared to the cost of (a part of) *either* component (as opposed to the lower component). As in [4], this runs the risk of charging many merging operations to a higher component, but by performing an accounting by tree decomposition level and using the fact that the decomposition only has h levels, we are able to show that our solution is still near-optimal even though we merge components much more aggressively than [4]. In this way, for each bag we construct one partition of its active terminals into a number of sets that is polynomial in $k + h + \frac{1}{\varepsilon} + \log n$, in a way that guarantees that this partition is a *refinement* of a near-optimal solution. That is, whenever we decide to place two terminals together in our partition, the near-optimal solution does the same. However, this solution does not necessarily conform to the resulting partitions, as two terminals of the same component might end up in different sets of the partition for a bag.

At this point an astute reader may be wondering that since we consider both the width k and the height h of the decomposition as parameters, we are effectively parameterizing by treedepth, rather than treewidth. This is correct, but we then go on to invoke a result of Bodlaender and Hagerup [6] which states that any tree decomposition can be rebalanced to have height $O(\log n)$ without severely increasing its width. Hence, the family of partitions we now have has size polynomial in $k + \frac{1}{\varepsilon} + \log n$. However, we are not done yet, since at this point we can only guarantee that our partitions are refinements of a near-optimal solution. To complete the algorithm, we work from this family of partitions to obtain a collection of partitions *conforming* to our near-optimal solutions using δ-nets (this is similar to the approach of [4]).

This leads to a running time of the form $(\log n)^{O(\frac{k^2}{\varepsilon})}n^{O(1)}$, which by standard arguments of parameterized complexity is in fact FPT and can be upper-bounded by a function of the form $2^{O(\frac{k^2}{\varepsilon}\log\frac{k}{\varepsilon})} \cdot n^{O(1)}$. To summarize, our high-level strategy is to show that the approach of [4] can be significantly improved when the input decomposition has small width and height, but then we observe that our new scheme is efficient enough in the height that even if we replace h by a bound that can be obtained for *any* graph, we still have an algorithm with an FPT running time, that is, significantly faster than that of [4].

Vertex Cover. For the parameterization by the vertex cover size, as mentioned we obtain an FPT exact algorithm with dependence $2^{O(k \log k)}$. A similar algorithm was recently independently obtained by Bodlaender et al. [8] via dynamic programming. However, our algorithm is significantly simpler, because our strategy is to show how to construct a tree decomposition and a collection of partitions Π_B such that we only need *one* partition of the active terminals for each bag. As a consequence, $p = O(n)$ and Theorem 5 implies the algorithm of Theorem 4, without the need to formulate a new dynamic program.

Our main result for this parameter is that under ETH the runtime dependence is asymptotically optimal. Note that this also implies that the runtime of the dynamic program given by Theorem 5 cannot be improved with regards to the dependence on the treewidth. To show this, we present a reduction from 3-SAT, where the goal is to compress an n-variable formula into a Steiner Forest instance such that the graph has vertex cover size $O(n/\log n)$. The intuition on why it is possible to achieve such a compression is the following: suppose we have an instance with vertex cover of size k and a demand between two vertices of the independent set. Then the simplest way to satisfy such a demand is to connect both vertices to a common neighbor in the vertex cover. This encodes a choice among k vertices, and hence it is sufficient to encode the assignment for $\log(k)$ binary variables. The strategy of our reduction is to set up some choice gadgets which allow us to encode the assignments to the original formula taking advantage of the fact that each choice can represent a logarithmic number of variables. Hence we can obtain a construction of slightly sub-linear ($O(n/\log n)$) size. We then of course need to add some verification gadgets, representing the clauses, to check that the formula is indeed satisfied. But even though the number of such gadgets is linear in n, we make sure that they form an independent set, and hence the total vertex cover size remains sufficiently small to obtain our lower bound. We note that this compression strategy is similar to techniques recently used to obtain slightly super-exponential lower bounds for vertex cover for other problems [28, 29], but the constructions we use are new and tailored to Steiner Forest.

Feedback Edge Set. For the parameterization by the size k of a feedback edge set, instead of relying on the dynamic program given by Theorem 5 we go an entirely different route in order to obtain the faster $2^{O(k)}n^{O(1)}$ time FPT algorithm of Theorem 3. First off, it is not hard to reduce the Steiner Forest problem to an instance in which all vertices have degree at least 2. We then consider paths with internal vertices of degree 2, with endpoints that are vertices incident to the feedback edge set or vertices of degree at least 3. We call these paths *topo-edges* and argue that there are only $O(k)$ of these. We then guess for which topo-edges the two endpoints lie in different components of the optimal Steiner Forest solution, which can be done in $2^{O(k)}$ time. If a topo-edge has both its endpoints in the same component of the optimum, we show that it can be easily handled. For the remaining topo-edges, we can decide which edges along the path do not belong to the optimal solution by a reduction to the polynomial-time solvable Min Cut problem.

1.2 Related work

Bateni et al. [4] show that one of the consequences of their XP approximation scheme is a PTAS for Steiner Forest on planar graphs, by using the common technique pioneered by Baker [2] of reducing this problem to graphs for which the treewidth is bounded as a function of ε. Because their algorithm is not FPT, their PTAS has a running time of the form $n^{f(\varepsilon)}$. By using our algorithm from Theorem 1 we can improve this runtime to $f(\varepsilon)n^{O(1)}$, i.e., we obtain on EPTAS for planar graphs. However, Eisenstat et al. [18] already showed that a $(1+\varepsilon)$-approximation algorithm with a runtime of $O(f(\varepsilon) \cdot n\log^3 n)$ exists for Steiner Forest on planar graphs. While they build on the work of Bateni et al. [4], and in particular also reduce to graphs of treewidth bounded as a function of ε, interestingly they do not obtain an EPAS parameterized by treewidth. Instead they use a different route and show that given a graph H of treewidth k, in $O(f(k,\varepsilon) \cdot n\log^2 n)$ time it is possible to compute a Steiner Forest solution in H whose cost is at most $\mathrm{cost}(F^\star) + \varepsilon\,\mathrm{cost}(H)$, i.e., there is an additive error that depends on the cost of H compared to the optimum solution $F^\star$. If the input graph G is planar, then a result by Borradaile et al. [9] implies that from G a so-called *banyan* [3, 32] can be computed, which is a subgraph

of G with cost bounded by $O(g(\varepsilon)\,\mathrm{cost}(F^\star))$, and which contains a near-optimal approximation of every Steiner forest (cf. [18, Lemma 2.1]). By applying the framework of Bateni et al. [4] on the banyan instead of the input graph, it is then possible to obtain a graph H of treewidth bounded by a function of ε, for which the algorithm of Eisenstat et al. [18] computes a $(1+O(\varepsilon))$-approximation for the input.

If it would be possible to compute a banyan for bounded treewidth graphs, then the algorithm of Eisenstat et al. [18] would also imply an EPAS for treewidth. However, to the best of our knowledge, and as explicitly stated by Bartal and Gottlieb [3], banyans are only known for planar graphs [9, 18], Euclidean metrics [32], and doubling metrics [3] (in fact, the latter are so-called *forest banyans*, which have weaker properties). Thus it is unclear how to obtain an EPAS for Steiner Forest parameterized by the treewidth via the algorithm of Eisenstat et al. [18]. We leave open whether a banyan exists for bounded treewidth graphs, which could give an alternative algorithm to the one given in Theorem 1. However, a further remark is that the cost of the banyan for planar graphs obtained by Borradaile et al. [9] has exponential dependence on $1/\varepsilon$, which implies a double exponential runtime dependence on $1/\varepsilon$ for the EPTAS for planar graphs. If a banyan can be obtained for bounded treewidth graphs by generalizing the techniques of Borradaile et al. [9] to minor-free graphs, then the resulting EPAS parameterized by treewidth would also have double exponential runtime in $1/\varepsilon$. In this case however, our EPAS given by Theorem 1 would be exponentially faster.

A different parameter that is often studied in the context of Steiner problems is the number $p = |R|$ of terminals. The classic result of Dreyfus and Wagner [15] presents an FPT algorithm for Steiner Tree with a runtime of $3^p n^{O(1)}$. For unweighted graphs, this was improved [5, 31] to $2^p n^{O(1)}$, while the fastest known algorithm for weighted graphs can compute the optimum in $(2+\varepsilon)^p n^{O(\sqrt{\frac{1}{\varepsilon}\log\frac{1}{\varepsilon}})}$ time [20] for any $\varepsilon > 0$. The algorithm of Dreyfus and Wagner [15] can be generalized to solve Steiner Forest in $2^{O(p)}n^{O(1)}$ time (cf. [12]). A somewhat dual parameter to the number of terminals is the number q of non-terminals (so-called *Steiner vertices*) in the optimum solution. For this parameter, a folklore result states that Steiner Tree (and thus also Steiner Forest) is W[2]-hard (cf. [14, 17]). However, an EPAS with a runtime of $2^{O(q^2/\varepsilon^4)}n^{O(1)}$ was shown to exist for Steiner Tree [17]. For Steiner Forest it is not hard to see that such an EPAS parameterized by q cannot exist unless P=NP (cf. [17]), but if c denotes the number of components of the optimum solution, there is an EPAS with a runtime of $(2c)^{O((q+c)^2/\varepsilon^4)}n^{O(1)}$ [17]. Similar results have been found for related Steiner problems in directed graphs [12]. For further results in the area of parameterized approximations, we refer to the survey [19].

2 Preliminaries

As mentioned, we assume the reader is familiar with the basics of parameterized complexity, such as the class FPT [14], and approximation algorithms such as a PTAS [36]. A *parameterized approximation scheme (PAS)* is an algorithm that computes a $(1+\varepsilon)$-approximation for a problem in $f(k,\varepsilon)n^{g(\varepsilon)}$ time for some functions f and g, while an an *efficient parameterized approximation scheme (EPAS)* is a $(1+\varepsilon)$-approximation algorithm running in time $f(k,\varepsilon)n^{O(1)}$ (that is, the running time is FPT in $k+\frac{1}{\varepsilon}$). The distinction between a PAS and an EPAS is similar to the one between a PTAS and an EPTAS.

By $w : E \to \mathbb{R}^+$ we denote an edge-weight function, so that the cost of a solution F to the Steiner Forest problem is $\mathrm{cost}(F) = \sum_{e\in E(F)} w(e)$. We will use $F^\star$ to denote an optimal solution, and for $\alpha \geq 1$ we will say that a solution F is α-approximate if $\mathrm{cost}(F) \leq \alpha\,\mathrm{cost}(F^\star)$. For $u, v \in V$ we use $\mathrm{dist}(u,v)$ to denote the shortest-path distance from u to v in G according to the weight function w.

Definition 6. *Given a graph $G = (V,E)$, a* tree decomposition *is a pair $(T, \{B_i\}_{i\in V(T)})$, where T is a tree and each node $i \in V(T)$ of the tree is associated with a* bag *$B_i \subseteq V$, with the following properties:*

1. *$\bigcup_{i\in V(T)} B_i = V$, i.e., all vertices of G are covered by the bags,*
2. *for every edge $uv \in E$ of G there exists a node $i \in V(T)$ of the tree for which $u, v \in B_i$, and*
3. *for every vertex $v \in V$ of G the nodes $\{i \in V(T) \mid v \in B_i\}$ of the tree for which the bags contain v induce a (connected) subtree of T.*

The width *of the tree decomposition is $\max_{i\in V(T)}\{|B_i| - 1\}$ and the* treewidth *of G is the minimum width over all its tree decompositions.*

A rooted tree decomposition is nice *if for every $i \in V(T)$ we have one of the following:*

1. *i has no children*[2] *(i is a* leaf node*),*

[2]here we do not demand the leaf nodes to be empty, as is often assumed for this definition.

2. *i has exactly two children i_1 and i_2 such that $B_i = B_{i_1} = B_{i_2}$ (i is a* join node*),*

3. *i has a single child i' where $B_i = B_{i'} \cup \{v\}$ for some $v \in V$ (i is an* introduce node*), or*

4. *i has a single child i' where $B_i = B_{i'} \setminus \{v\}$ for some $v \in V$ (i is a* forget node*).*

Given a rooted tree decomposition T of a graph G, for a node u of T let B be the bag associated with it. Then V_B is the set of vertices of all bags in the subtree rooted at u. The set $A_B \subseteq R$ denotes the *active terminals* of the bag B: for any demand pair $\{s,t\} \in D$, if $s \in V_B$ and $t \notin V_B$ then $s \in A_B$. For any Steiner Forest solution F, if a connected component C of F contains an active terminal, then we say that C is an *active component* for B. For a fixed solution F, we denote the set of all active components for B by $\mathcal{C}_B$. Note that every active component must intersect the bag B.

If for every bag B a set of partitions Π_B of A_B is given, a Steiner Forest solution F is *conforming* to all Π_B, if for each bag B there exists a partition $\pi \in \Pi_B$ such that any two active terminals in A_B are in the same set $S \in \pi$ if and only if they are part of the same active component C of F, i.e., $S \subseteq V(C)$ and $S' \cap V(C) = \emptyset$ for any $S' \in \pi$ with $S' \neq S$ (note that this implies $|\pi| \leq |B|$). One technicality of Theorem 5 is that the algorithm needs a nice tree decomposition as input, for which the terminals only appear in bags that are leaf nodes of the decomposition. Given any tree decomposition, these conditions are not hard to meet (cf. [4, Lemma 6]). However, for our algorithms we are going to rely on tree decompositions with certain additional properties. Hence we will need to revisit the conditions needed for the algorithm of Theorem 5 when using it for our purposes.

We will also consider the following parameters: The *treedepth* of a graph G can be defined recursively as follows: (i) the treedepth of K_1 is 1 (ii) the treedepth of a disconnected graph is the maximum of the treedepth of any of its components (iii) the treedepth of a connected graph G is $1 + \min_{v \in V(G)} \mathrm{td}(G - v)$. A *feedback vertex set* is a set of vertices whose removal leaves a forest. A *vertex cover* is a set of vertices such that its removal leaves an edge-less graph. A *feedback edge set* is a set of edges whose removal leaves a forest. In a connected graph with n vertices and m edges the minimum feedback edge set always has size $m - n + 1$.

As part of our approximation algorithm we will use the notion of δ-nets, defined as follows. A well-known fact is that a δ-net exists for any metric and any $\delta \geq 0$, and it can be constructed greedily in polynomial time.

Definition 7. *Given a metric (X, dist), a* δ-net *is a subset $N \subseteq X$ of points, such that*

1. *any two net points $u, v \in N$ are far from one another, i.e., $\mathrm{dist}(u,v) > \delta$, and*

2. *for any node $u \in X$ there is some net point $v \in N$ close by, i.e., $\mathrm{dist}(u,v) \leq \delta$.*

3 An efficient parameterized approximation scheme for treewidth

In this section we describe the main result of this paper which is an EPAS for Steiner Forest parameterized by treewidth. We begin by giving two preliminary tools (Lemma 8 and Lemma 9) which facilitate the algorithm by ensuring that the given tree decomposition has logarithmic height and that the instance has aspect ratio (ratio of the weights of the heaviest over the lightest edge) bounded by a polynomial in n.

We then go on to subsection 3.1 where we introduce a second parameter, the height h of the decomposition. Our goal is to fix an almost-optimal solution F_ε and describe an algorithm that produces a partition ζ_B of the active terminals for each bag B of the decomposition, where ζ_B is *a refinement* of the partition implied by F_ε (Lemma 10). In other words, we seek a partition ζ_B of A_B such that if two terminals t_1, t_2 are in the same set of ζ_B, then they are also in the same component of F_ε. Of course, it is trivial to achieve this by giving a ζ_B where each active terminal is in its own set, so the interesting part here is how we group terminals together in a way that in the end allows us to bound $|\zeta_B|$ by a polynomial of $k + h + \frac{1}{\varepsilon} + \log n$, while still ensuring that F_ε is almost optimal.

The partition ζ_B of Lemma 10 is not yet conforming, because two terminals which are in distinct sets of ζ_B may still be in the same component of F_ε, and thus we cannot apply Theorem 5 at this point. Therefore in subsection 3.2, given ζ_B we focus on how to obtain every possible partition of the set of active terminals, which could be conforming with an almost-optimal solution. By an appropriate use of δ-nets, similar to [4], we are able to "guess" (that is, brute-force) a choice of a small number of net points per active component. Since the number of choices for each point is at most $|\zeta_B|$ and we choose roughly

$O(k^2/\epsilon)$ points in total, the total number of produced partitions (and hence the running time given by Theorem 5) is of the form $(\log n + k + \frac{1}{\epsilon})^{O(k^2/\epsilon)} n^{O(1)}$, which is FPT.

Let us now recall a result of Bodlaender and Hagerup [6] which states that a tree decomposition of logarithmic height can always be obtained.

Lemma 8 ([6]). *Given a tree decomposition of width k of a graph G on n vertices, there is a polynomial time algorithm computing a nice tree decomposition of G of width $O(k)$ and height $O(k \log n)$.*

Proof. It is shown in [6] that any tree decomposition of width k can be transformed in polynomial time into a tree decomposition of width $O(k)$ and height $O(\log n)$ where the tree of the decomposition has maximum degree 3. It is now not hard to make this decomposition nice by replacing all nodes with two children by Join nodes, and by inserting between any node and its parent a sequence of $O(k)$ new nodes so that the symmetric difference between any node and its parent contains at most one vertex. □

We also need to reduce the aspect ratio of the given graph to a polynomial. This can be done using a standard technique, where however we need to make sure that the treewidth of the given graph remains bounded. Note that the aspect ratio of the resulting graph G' in the following lemma is polynomially bounded in the size of the original graph, but not necessarily in the size of G' (because G' may have significantly fewer vertices).

Lemma 9. *Given $\varepsilon > 0$, an instance of Steiner Forest on a graph G with n vertices, and a (nice) tree decomposition T of width k and height h for G, in polynomial time we can compute an instance on a graph G' with at most n vertices and a (nice) tree decomposition T' of width at most k and height h for G', such that the ratio of the longest to the shortest edge in G' is at most $2n/\varepsilon$, and any α-approximation for G' can be converted into an $(\alpha + \varepsilon)$-approximation for G.*

Proof. The first step is to compute a 2-approximation F_2 for Steiner Forest in G, using the polynomial time algorithm of [1]. The new graph G' is obtained from G by first removing all edges of length more than $\mathrm{cost}(F_2)$ and then contracting every edge of length less than $\frac{\varepsilon}{2n}\mathrm{cost}(F_2)$, where n is the number of vertices of G. If a contracted edge was incident to a terminal, then the new vertex is declared a terminal and the demands are updated correspondingly (note that this may introduce trivial demands from the new terminal to itself if a demand pair is connected by a path of edges being contracted). We modify the tree decomposition T to obtain T' as follows: whenever we contract the endpoints of an edge v_1v_2 into a new vertex w, we replace all occurrences of v_1 and v_2 in T by w. It is not hard to see that this keeps a valid decomposition of the same height and can only decrease the width. Furthermore, if the original decomposition was nice, the new decomposition can easily be made nice, if we contract every bag B with a unique child B' whenever $B = B'$ (which were introduce or forget nodes previously). Also, clearly the ratio between longest and shortest edge in G' is at most $2n/\varepsilon$.

It remains to show that an α-approximate solution F'_α in G' is not distorted by much when converting it from G' to G. Starting with F'_α the conversion is simply done by iteratively uncontracting those edges that were contracted to obtain G' from G: if the solution becomes infeasible after uncontracting some edge e we just add it to the solution to make it feasible again. Let F denote the solution obtained for G from F'_α, and note that less than n edges are added to F'_α in this process, as F is a forest in G. This means that $\mathrm{cost}(F) < \mathrm{cost}(F'_\alpha) + \frac{\varepsilon}{2}\mathrm{cost}(F_2)$ since every contracted edge has length less than $\frac{\varepsilon}{2n}\mathrm{cost}(F_2)$. Now consider an optimum solution $F^\star$ in G. It can be converted into a solution of cost at most $\mathrm{cost}(F^\star)$ in G' by contracting all edges of length less than $\frac{\varepsilon}{2n}\mathrm{cost}(F_2)$, since $F^\star$ cannot contain any of the removed edges of length more than $\mathrm{cost}(F_2) \geq \mathrm{cost}(F^\star)$. Thus the optimum of G' has cost at most $\mathrm{cost}(F^\star)$, and because F'_α is an α-approximation in G' we get $\mathrm{cost}(F'_\alpha) \leq \alpha\,\mathrm{cost}(F^\star)$. At the same time, $\mathrm{cost}(F_2) \leq 2\,\mathrm{cost}(F^\star)$, which together with the previous inequality gives $\mathrm{cost}(F) < \alpha\,\mathrm{cost}(F^\star) + 2\frac{\varepsilon}{2}\mathrm{cost}(F^\star) = (\alpha + \varepsilon)\,\mathrm{cost}(F^\star)$, which concludes the proof. □

For simplicity, in the following we will scale the edge lengths of any given graph so that the shortest edge has length 1. In particular, after applying Lemma 9, the longest edge has length at most $2n/\varepsilon$.

3.1 Tree decompositions with bounded height

In this section we informally assume that the height h of the given tree decomposition is bounded as well as the width k. Our aim is to prove the following statement, where we restrict ourselves to input graphs of polynomial aspect ratio, which we may do according to Lemma 9 (keeping in mind that n is the number of vertices of the original input graph).

Lemma 10. *Let an instance of* Steiner Forest *on a graph G with at most n vertices be given together with a tree decomposition T of width k and height h for G. For any $\varepsilon > 0$, if the ratio between the longest and shortest edge of G is at most $2n/\varepsilon$, then there exists a $(1+\varepsilon)$-approximation F_ε with the following properties. There exists a polynomial time algorithm, which for every bag B of T outputs a partition ζ_B of the active terminals A_B, such that each set of ζ_B belongs to the same component of F_ε and $|\zeta_B| = O(\frac{k^4h^2}{\varepsilon^2}\log\frac{n}{\varepsilon})$.*

To prove Lemma 10 we first identify the solution F_ε, after which we will show how to compute the partitions ζ_B.

3.1.1 A near-optimal solution

The high-level idea to obtain a $(1+\varepsilon)$-approximate solution F_ε is to connect components of the optimum solution $F^\star$ that lie very close to each other. In particular, if the distance between two components C and C' of $F^\star$ is of the form $f(k,h,\varepsilon)\,\mathrm{cost}(C)$ for some small enough function f, then we may hope to add a shortest path between C and C' and charge this additional cost to C, in order to obtain a $(1+\varepsilon)$-approximation. Unfortunately, this approach is not viable, since the number of components that are very close to C may be very large, meaning that the function f in the distance bound would have to linearly depend on the number of vertices in order to result in a $(1+\varepsilon)$-approximation. This in turn would mean that the size of the partition ζ_B would depend polynomially on the number of vertices, making it unsuitable for an FPT time algorithm. This issue lies at the heart of the problem and is the reason for why it is non-trivial to obtain an approximation scheme parameterized by the treewidth. To get around this issue we will measure the distance between components using a modified cost function, which we define next.

Given a bag B of the rooted tree decomposition T, we denote by T_B the subtree of T rooted at the node associated with B, and by $G_B = G[V_B]$ the graph induced by the vertices V_B lying in bags of T_B. We also define the graph $G_B^\downarrow \subseteq G_B$ as the graph spanned by all edges of G_B, except those induced by B, i.e., the edge set of $G_B^\downarrow$ is

$$E(G_B^\downarrow) = \{uv \in E(G_B) \mid u \notin B \vee v \notin B\}.$$

The cost of a component C of some Steiner Forest solution restricted to $G_B^\downarrow$ only counts the edge weights of C in $G_B^\downarrow$, and is denoted by

$$\mathrm{cost}_B^\downarrow(C) = \sum_{e \in E(C) \cap E(G_B^\downarrow)} w(e).$$

Based on these definitions, we fix an optimal solution $F^\star$ and construct a solution F_ε by initially setting $F_\varepsilon = F^\star$, and then connecting components by exhaustively applying the following rule, where we say that two components C and C' *share* a bag B if $V(C) \cap B \neq \emptyset$ and $V(C') \cap B \neq \emptyset$:

Rule 1: if C, C' are components of $F^\star$ sharing a bag B with $\mathrm{dist}(C,C') \leq \frac{\varepsilon}{kh} \cdot \mathrm{cost}_B^\downarrow(C)$ but C and C' are in different components of F_ε, then add a shortest path of length $\mathrm{dist}(C,C')$ between C and C' to the solution F_ε.

Lemma 11. *The cost of the solution F_ε obtained by Rule 1 from $F^\star$ is at most $(1+\varepsilon)\,\mathrm{cost}(F^\star)$.*

Proof. It suffices to prove that the cost of all paths added to $F^\star$ in order to obtain F_ε according to Rule 1 is at most $\varepsilon \cdot \mathrm{cost}(F^\star)$. For this we use a charging scheme that charges new paths to components of $F^\star$. In particular, we charge a path of length $\mathrm{dist}(C,C') \leq \frac{\varepsilon}{kh} \cdot \mathrm{cost}_B^\downarrow(C)$ to component C.

Fix a component C of $F^\star$ and a bag B with $V(C) \cap B \neq \emptyset$. We define $\mathrm{charge}(C,B)$ to be the cost we charge to C for operations involving other components of $F^\star$ that share B. It is not hard to see that $\mathrm{charge}(C,B) \leq \frac{\varepsilon}{h} \cdot \mathrm{cost}_B^\downarrow(C)$, because there are at most k other components of $F^\star$ that share B.

For $\ell \in \{0,\ldots,h-1\}$, let $\mathcal{B}_\ell$ be the set of bags of the tree decomposition that appear at distance exactly ℓ from the root, i.e., they lie on level ℓ of the tree. We now observe that

$$\sum_{B \in \mathcal{B}_\ell} \mathrm{charge}(C,B) \leq \sum_{B \in \mathcal{B}_\ell} \frac{\varepsilon}{h} \cdot \mathrm{cost}_B^\downarrow(C) \leq \frac{\varepsilon}{h}\,\mathrm{cost}(C),$$

where the last inequality follows because if we have two bags $B, B' \in \mathcal{B}_\ell$, then $E(G_B^\downarrow) \cap E(G_{B'}^\downarrow) = \emptyset$: note that every edge of $E(G_B^\downarrow)$ must be incident on a vertex v that appears in a descendant of B, but not

in B. By the properties of tree decompositions, notably by the fact that B is a separator of G, v cannot appear in B' or any of its descendants. Therefore none of its incident edges are contained in $E(G^{\downarrow}_{B'})$. Because $\sum_{B\in\mathcal{B}_\ell} \mathrm{cost}^{\downarrow}_B(C)$ is the sum of costs of C over disjoint sets of edges, the sum is a lower bound on the total cost of C.

To conclude, we observe that the total charge of C is

$$\mathrm{charge}(C) \leq \sum_{\ell=0}^{h-1} \sum_{B\in\mathcal{B}_\ell} \mathrm{charge}(C,B) \leq \varepsilon\,\mathrm{cost}(C).$$

Therefore, summing over all components of $F^\star$, the total cost of the edges we have added according to Rule 1 is at most $\varepsilon \cdot \mathrm{cost}(F^\star)$. □

3.1.2 Partitioning active terminals

We are now ready to prove Lemma 10 for the near-optimal solution F_ε constructed above, for which we will compute the partitions ζ_B for all bags B. We will use the following two claims for the active terminals A_B of the given bag B.

Claim 12. *If there exist $t_1, t_2 \in A_B$ such that $\mathrm{dist}(t_1,t_2) \leq \frac{\varepsilon}{kh}\mathrm{dist}(t_1,B)$, then t_1, t_2 are in the same component of F_ε.*

Proof. Let C_1 be the component that contains t_1 in the optimal solution $F^\star$ from which F_ε is constructed. We observe that $\mathrm{cost}^{\downarrow}_B(C_1) \geq \mathrm{dist}(t_1,B)$, because C_1 must contain a path from t_1 to B (as t_1 is active) and all the edges of this path are contained in $E(G^{\downarrow}_B)$. If t_2 is contained in C_2 in $F^\star$ we therefore have, $\mathrm{dist}(C_1,C_2) \leq \mathrm{dist}(t_1,t_2) \leq \frac{\varepsilon}{kh}\mathrm{dist}(t_1,B) \leq \frac{\varepsilon}{kh}\mathrm{cost}^{\downarrow}_B(C_1)$. Since C_2 must also intersect B, Rule 1 implies that C_1 and C_2 are contained in the same component of F_ε. □

Claim 13. *Let $A \subseteq A_B$ and $d \geq 0$ be such that (i) there exists $b \in B$ such that for all $t \in A$ we have $\mathrm{dist}(t,B) = \mathrm{dist}(t,b)$ and $d \leq \mathrm{dist}(t,B) \leq 2d$ (ii) for all distinct $t,t' \in A$ we have $\mathrm{dist}(t,t') > \frac{\varepsilon}{kh}d$ (iii) $|A| \geq \frac{8k^2(k+1)h^2}{\varepsilon^2}$. Then, there exists a component of F_ε that contains all terminals of A.*

Proof. Consider an active component C for B of the optimum solution $F^\star$. We claim that $\mathrm{cost}^{\downarrow}_B(C) \geq |V(C)\cap A| \cdot \frac{\varepsilon}{2kh}d$. To see this, let $C^{\downarrow}_1, \ldots, C^{\downarrow}_\ell$ be the components of C when restricting C to $G^{\downarrow}_B$. Each component $C^{\downarrow}_i$ is a Steiner tree for the terminals in $V(C^{\downarrow}_i)\cap A$. Now consider a minimum spanning tree U on the metric closure derived from $G^{\downarrow}$ for this vertex set $V(C^{\downarrow}_i)\cap A$. It is well known that such a minimum spanning tree is a $2(1-\frac{1}{p})$-approximation of the optimum Steiner tree [27] on p terminals, and thus $\mathrm{cost}(C^{\downarrow}_i) \geq \frac{1}{2(1-\frac{1}{p})}\mathrm{cost}(U)$ where $p = |V(C^{\downarrow}_i)\cap A|$. The distance between any two terminals of $V(C^{\downarrow}_i)\cap A$ in the given graph G is more than $\frac{\varepsilon}{kh}d$ by property (ii) of the claim, and because the distance between such terminals can only be more in $G^{\downarrow}$, every edge of U has cost more than $\frac{\varepsilon}{kh}d$. This means that $\mathrm{cost}(U) \geq (p-1)\cdot\frac{\varepsilon}{kh}d$, and we therefore get $\mathrm{cost}(C^{\downarrow}_i) \geq \frac{p-1}{2(1-\frac{1}{p})}\cdot\frac{\varepsilon}{kh}d = \frac{p}{2}\cdot\frac{\varepsilon}{kh}d$. Summing over all (vertex disjoint) components $C^{\downarrow}_i$ we obtain the claimed inequality $\mathrm{cost}^{\downarrow}_B(C) \geq |V(C)\cap A|\cdot\frac{\varepsilon}{2kh}d$.

Because each terminal of A belongs to an active component of $F^\star$, of which there are at most $k+1$, there must exist an active component C with $|V(C)\cap A| \geq \frac{|A|}{k+1}$, which by the above inequality and property (iii) of the claim gives $\mathrm{cost}^{\downarrow}_B(C) \geq \frac{|A|}{k+1}\cdot\frac{\varepsilon}{2kh}d \geq \frac{kh}{\varepsilon}\cdot 4d$. Now note that by property (i), for all $t,t' \in A$ we have $\mathrm{dist}(t,t') \leq 4d$, as we can use a path through b. So, if C' is any of the other active components of $F^\star$ also containing a terminal of A, we have $\mathrm{dist}(C,C') \leq 4d$. We therefore obtain $\mathrm{dist}(C,C') \leq \frac{\varepsilon}{kh}\mathrm{cost}^{\downarrow}_B(C)$, and according to Rule 1, C and C' are part of the same component of F_ε. In other words, all active terminals of A are in components of $F^\star$ that lie in the one component of F_ε containing C. □

Intuitively, Claim 12 allows us to place terminals of A which are very close to each other into the same set of the partition ζ_B, as placing one terminal in a component forces the placement of the other. Thanks to this claim we can work with an appropriate net. If we find a large collection of such net points which also are roughly the same distance from the bag and closest to the same vertex of the bag, Claim 13 allows us to group them all together in the partition ζ_B. Armed with these tools, we can now prove the main lemma.

Proof of Lemma 10. To compute the partition ζ_B in polynomial time, we first partition the active terminals $A_B \cap B$ contained in the bag B. For this we simply add a set $\{t\}$ for each $t \in A_B \cap B$ to ζ_B, which adds at most $|B| \leq k+1$ sets to ζ_B. Let now $A = A_B \setminus B$ be the remaining active terminals.

To partition A, let $d = \min_{t \in A_B \setminus B} \operatorname{dist}(t, B)$ and $D = \max_{t \in A_B \setminus B} \operatorname{dist}(t, B)$ be the minimum and maximum distances of these active terminals from the bag B. Then partition $A_B \setminus B$ into $|B| \leq k+1$ sets $A_1, A_2, \ldots, A_{|B|}$, depending on the vertex of B that is closest to each $t \in A$ (breaking ties arbitrarily). That is, for each A_i there exists $b \in B$ such that for all $t \in A_i$ we have $\operatorname{dist}(t, B) = \operatorname{dist}(t, b)$. Consider now a set A_i and further partition it into $r = \lceil \log_2 \frac{D}{d} \rceil$ sets $A_{i,0}, A_{i,1}, \ldots, A_{i,r-1}$, where $A_{i,j}$ contains all $t \in A_i$ such that $\operatorname{dist}(t, B) \in [2^j d, 2^{j+1} d)$. Now (greedily) compute an $(\frac{\varepsilon}{kh} 2^j d)$-net $N_{i,j}$ of $A_{i,j}$. We observe that $N_{i,j}$ satisfies the first two conditions of Claim 13 for $2^j d$, so if $|N_{i,j}| \geq \frac{8k^2(k+1)h^2}{\varepsilon^2}$, then we add $A_{i,j}$ as a set of our partition ζ_B, remove the terminals of $N_{i,j}$ from A and continue the algorithm for the remaining terminals. Repeat the previous step for all i, j for which $N_{i,j}$ is sufficiently large. This contributes at most $(k+1)\lceil \log \frac{D}{d} \rceil$ sets to ζ_B.

Suppose now that we are left with a set of terminals A such that the procedure above fails to construct a sufficiently large net $N_{i,j}$ to apply Claim 13. For every index pair i, j, each remaining terminal $t \in A_{i,j}$ is close enough to some net point $t' \in N_{i,j}$ such that we can apply Claim 12. We therefore create a set in the partition ζ_B for each $t' \in N_{i,j}$, placing into such a set those terminals of $A_{i,j}$ that are closest to t' (breaking ties arbitrarily). Since we cannot apply Claim 13 to the remaining sets $A_{i,j}$, each of the at most $(k+1)\lceil \log \frac{D}{d} \rceil$ nets $N_{i,j}$ has size less than $\frac{8k^2(k+1)h^2}{\varepsilon^2}$, which implies $|\zeta_B| \leq O(\frac{k^4 h^2}{\varepsilon^2} \log \frac{D}{d})$.

Clearly the above procedure can be implemented in polynomial time, and the fact that every set of ζ_B is contained in the same component of F_ε follows from Claim 12 and Claim 13. Finally, any path in a graph with at most n vertices has less than n edges, so that $\frac{D}{d} < 2n^2/\varepsilon$, given that the ratio of the longest to the shortest edge is $2n/\varepsilon$ (note that $d > 0$ by definition). Hence the claimed bound of $|\zeta_B| \leq O(\frac{k^4 h^2}{\varepsilon^2} \log \frac{n}{\varepsilon})$ follows. □

3.2 Tree decompositions with logarithmic height

Given a tree decomposition T of logarithmic height, using Lemma 10 we are ready to compute a set of partitions Π_B of FPT size for each bag B, such that a near-optimal solution conforms to Π_B. In particular, by Lemma 8 we may assume that the height of T is $h = O(k \log n)$, which means that the bound on ζ_B in Lemma 10 translates to $O(\frac{k^6}{\varepsilon^2} \log^3 \frac{n}{\varepsilon})$. As in the previous section, we need to apply Lemma 9 in order to bound the aspect ratio of the graph, so that n denotes the number of vertices of the original input graph, while now the graph G has at most n vertices but the ratio between the longest and shortest edge is at most $2n/\varepsilon$. We begin by describing how to obtain the near-optimal solution, after which we will identify the partition sets Π_B.

3.2.1 A near-optimal solution

Bateni et al. [4] construct a near-optimal solution by modifying the optimum. We will use similar techniques to obtain our near-optimal solution, but we construct it by instead modifying the $(1+\varepsilon)$-approximate solution F_ε given by Lemma 10. In particular, we construct a near-optimal $(1+\varepsilon)^2$-approximation $\widetilde{F}_\varepsilon$ from F_ε. The main idea to obtain $\widetilde{F}_\varepsilon$ is to connect components of F_ε if they are very close to one another. As before however, doing this naively would incur too much cost for the additional connections.

To make sure that the cost incurred by connecting components of F_ε is not too large, [4] introduced a partial order on the components based on the structure of a given rooted tree decomposition T. Let C_1, C_2 be two components of F_ε that share a bag B of T, i.e., $V(C_1) \cap B \neq \emptyset$ and $V(C_2) \cap B \neq \emptyset$. Since C_1 and C_2 are connected subgraphs of the input graph, a basic property of tree decompositions implies that there are (connected) subtrees T_1 and T_2 of T induced by the respective bags containing vertices of C_1 and C_2. Because these components both contain vertices of B, the node associated with B is part of both T_1 and T_2, and therefore the roots of both subtrees lie on the path from this node to the root of T. This defines an order on C_1 and C_2, and we write $C_1 \leq C_2$ if the root of T_1 is farther from the root of T than the root of T_2 is. This order is defined for any two components of F_ε that share a bag, and thus we obtain a partial order on the components of F_ε, where any components that do not share a bag are incomparable.

Using the defined order, Bateni et al. [4] connect components of the optimum solution that are very close to each other. In order to obtain smaller partition sets, we modify the distance bound used in this procedure compared to Bateni et al. [4]. In particular, for any value $x > 0$, let $\lfloor x \rfloor_2 = 2^{\lfloor \log_2 x \rfloor}$ denote the

largest power of 2 that is at most x. Now, starting with $\widetilde{F}_\varepsilon = F_\varepsilon$ we connect components by exhaustively applying the following rule:

Rule 2: if C, C' are components of F_ε with $C \leq C'$ and $\operatorname{dist}(C, C') \leq \frac{\varepsilon}{k}\lfloor \operatorname{cost}(C)\rfloor_2$ but C and C' lie in different components of $\widetilde{F}_\varepsilon$, then add a shortest path of length $\operatorname{dist}(C, C')$ between C and C' to the solution $\widetilde{F}_\varepsilon$.

A crucial but subtle observation is that for a component C of F_ε there can be many components $C' \leq C$ at distance at most $\frac{\varepsilon}{k}\lfloor \operatorname{cost}(C)\rfloor_2$ to C, which however are not connected to C in the resulting solution $\widetilde{F}_\varepsilon$ according to Rule 2. This makes it non-trivial to find small partition sets Π_B. Contrary to this however, an important property of the order on the components is that for any component C of F_ε, there are at most k other components C' for which $C \leq C'$, as we will argue for the following lemma to bound the cost of $\widetilde{F}_\varepsilon$. In particular, the lemma implies that $\widetilde{F}_\varepsilon$ is a near-optimal $(1+\varepsilon)^2$-approximation, given that F_ε is a $(1+\varepsilon)$-approximation.

Lemma 14. *The cost of the solution $\widetilde{F}_\varepsilon$ obtained by Rule 2 from F_ε is at most $(1+\varepsilon)\operatorname{cost}(F_\varepsilon)$.*

Proof. Consider a component C of F_ε and the highest (closest to the root) node of T for which the bag B contains a vertex of C. Any component C' with $C \leq C'$ also intersects B, and as this bag has size at most $k+1$, there can be at most k such components C'. As a consequence, we can charge the additional cost of connecting a component C with components C' for which $C \leq C'$ to the cost of C. In particular, if $\mathcal{C}$ denotes the set of all components of F_ε, then the cost incurred by connecting components according to Rule 2 is at most

$$\sum_{C\in\mathcal{C}} \sum_{C'\in\mathcal{C}:C\leq C'} \frac{\varepsilon}{k}\lfloor \operatorname{cost}(C)\rfloor_2 \leq \sum_{C\in\mathcal{C}} \sum_{C'\in\mathcal{C}:C\leq C'} \frac{\varepsilon}{k}\operatorname{cost}(C) \leq \sum_{C\in\mathcal{C}} \varepsilon \operatorname{cost}(C) = \varepsilon \operatorname{cost}(F_\varepsilon).$$

Thus adding the cost of connecting components of F_ε according to Rule 2, the cost of the resulting solution is $\operatorname{cost}(\widetilde{F}_\varepsilon) \leq \operatorname{cost}(F_\varepsilon) + \varepsilon \operatorname{cost}(F_\varepsilon) = (1+\varepsilon)\operatorname{cost}(F_\varepsilon)$. □

3.2.2 Partitioning active terminals

Given the construction of the $(1+\varepsilon)^2$-approximate solution $\widetilde{F}_\varepsilon$ above, the next step is to find a set of partitions Π_B of the active terminals A_B for each bag B, such that $\widetilde{F}_\varepsilon$ conforms with all sets Π_B. In the following, fix a bag B of the given tree decomposition T. The technique used by Bateni et al. [4] is to guess a small net for each active component of bag B,[3] so that every terminal of A_B close to a net point must be part of the same component in the approximate solution, after taking the order on the active components as defined previously into account. Next we choose a net on the terminals of each active component and bound its size.

Lemma 15. *Let $N \subseteq A_B \cap C$ be an $\frac{\varepsilon}{k}\lfloor \operatorname{cost}(C)\rfloor_2$-net of the metric induced by the active terminals of some active component C. The size of the net can be bounded by $|N| \leq \lfloor 4k/\varepsilon \rfloor$.*[4]

Proof. Let U be a minimum spanning tree of the metric closure of N. It is well known that a minimum spanning tree is a $2(1-\frac{1}{p})$-approximation to an optimum Steiner tree [27] on p terminals, and thus we have $\operatorname{cost}(C) \geq \frac{1}{2(1-\frac{1}{|N|})} \cdot \operatorname{cost}(U)$, as C in particular is a Steiner tree for N. The distance between any pair of net points in N is more than $\frac{\varepsilon}{k}\lfloor \operatorname{cost}(C)\rfloor_2 \geq \frac{\varepsilon}{2k}\operatorname{cost}(C)$, and given that the spanning tree U has $|N|-1$ edges we get $\operatorname{cost}(U) > \frac{\varepsilon}{2k}\operatorname{cost}(C)(|N|-1)$. Putting these two inequalities together we get $\operatorname{cost}(C) > \frac{\varepsilon(|N|-1)}{4k(1-\frac{1}{|N|})}\operatorname{cost}(C) = \frac{\varepsilon|N|}{4k}\operatorname{cost}(C)$, which implies $|N| \leq \lfloor 4k/\varepsilon \rfloor$ as $|N|$ is an integer. □

Following the algorithm of Bateni et al. [4], the next step would be to guess such an $\frac{\varepsilon}{k}\lfloor \operatorname{cost}(C)\rfloor_2$-net for each of the at most $k+1$ active components C of the bag B. By Lemma 15, the total number of net points for these at most $k+1$ nets is at most $\lfloor 4k/\varepsilon \rfloor(k+1) = O(k^2/\varepsilon)$. Since however there may be up to n active terminals, guessing these nets for all active components can result in $n^{O(k^2/\varepsilon)}$ many possible choices, which leads to an XP time algorithm. To circumvent this, we instead consider the partition ζ_B of the active terminals as given by Lemma 10, and guess which of the sets of ζ_B contains a net point. We will argue that since the size of ζ_B is $O(\frac{k^6}{\varepsilon^2}\log^3\frac{n}{\varepsilon})$ there are only $(\frac{k}{\varepsilon}\log\frac{n}{\varepsilon})^{O(k^2/\varepsilon)}$ possibilities, leading to a faster algorithm.

[3] Bateni et al. [4] refer to these nets as *groups*.

[4] A slightly worse bound follows from [4, Lemma 19].

More concretely, to compute a set of partitions Π_B that $\widetilde{F}_\varepsilon$ conforms to, our algorithm considers every sequence $((S_1,\delta_1),(S_2,\delta_2),\dots,(S_\ell,\delta_\ell),\rho)$ of at most $k+1$ pairs (S_j,δ_j) and partitions ρ of the index set $\{1,\dots,\ell\}$, where each S_j is a subset of the parts of ζ_B such that $|S_j| \leq \lfloor 4k/\varepsilon \rfloor$, and $\delta_j \in \{2^q \mid q \in \mathbb{N}_0 \wedge 0 \leq q \leq \log_2(2n^2/\varepsilon)\}$ is an integer power of 2 between 1 and $2n^2/\varepsilon$, where n is the number of vertices of the original input graph in accordance with Lemma 9. From every such sequence, the algorithm attempts to construct a partition of the active terminals, and if it succeeds adds it to the set Π_B. As we will show, in this process the algorithm will successfully construct one partition π of A_B that $\widetilde{F}_\varepsilon$ conforms to.

Before describing how a partition of the active terminals arises from such a sequence, we bound the number of these sequences, which determines the running time. By Lemma 10, $|\zeta_B| = O(\frac{k^6}{\varepsilon^2}\log^3\frac{n}{\varepsilon})$ if the tree decomposition T has logarithmic height, so that there are at most $\binom{|\zeta_B|}{\lfloor 4k/\varepsilon \rfloor} = (\frac{k}{\varepsilon}\log\frac{n}{\varepsilon})^{O(k/\varepsilon)}$ possible choices for each S_j. Clearly there are $O(\log\frac{n}{\varepsilon})$ choices for each δ_j, and $\ell^\ell = k^{O(k)}$ possible partitions ρ, given that $\ell \leq k+1$. Since a sequence contains ℓ sets S_j, the total number of sequences is bounded by $(\frac{k}{\varepsilon}\log\frac{n}{\varepsilon})^{O(k^2/\varepsilon)}$.

Each sequence may give rise to a partition $\pi \in \Pi_B$ of the active terminals as follows. First, let $\pi = \{Y_1,\dots,Y_{|\rho|}\}$, i.e., π has the same number of sets as the partition ρ. Let $U_j = \bigcup_{U\in S_j} U$ denote the set of active terminals in S_j, and let $\rho(j)$ be the part of ρ containing j. We distinguish between active terminals $t \in A_B$ that lie in some set U_j and those that do not:

- if $t \in U_j$ for some $j \in [\ell]$ then $t \in Y_{\rho(j)}$ (i.e., $U_j \subseteq Y_{\rho(j)}$), and
- otherwise, if $p_t \in \{1,\dots,\ell\}$ denotes the smallest index for which $\operatorname{dist}(t,U_{p_t}) \leq \frac{\varepsilon}{k}\delta_{p_t}$, then $t \in Y_{\rho(p_t)}$.

If this π is a partition of A_B we add π to Π_B, and otherwise we dismiss the current sequence. Clearly π can be constructed in polynomial time, given a sequence.

Lemma 16. *The $(1+\varepsilon)^2$-approximate solution $\widetilde{F}_\varepsilon$ conforms to the set Π_B of partitions constructed above.*

Proof. Consider the $(1+\varepsilon)$-approximate solution F_ε of Lemma 10 from which $\widetilde{F}_\varepsilon$ is constructed according to Rule 2, and the partition ζ_B of A_B as given by Lemma 10. Let the active components of F_ε be $C_1,\dots,C_\ell$ indexed according to their order, i.e., $C_j \leq C_{j'}$ if and only if $j \leq j'$. For each active component C_j we fix an $\frac{\varepsilon}{k}\lfloor\operatorname{cost}(C_j)\rfloor_2$-net N_j of size at most $\lfloor 4k/\varepsilon \rfloor$ according to Lemma 15. Now, consider the sequence $((S_1,\delta_1),(S_2,\delta_2),\dots,(S_\ell,\delta_\ell),\rho)$, where

- S_j contains exactly those sets of ζ_B that contain at least one net point of N_j,
- $\delta_j = \lfloor\operatorname{cost}(C_j)\rfloor_2$, and
- ρ is the partition of the index set corresponding to the components of $\widetilde{F}_\varepsilon$, i.e., $\rho(j) = \rho(j')$ if and only if C_j and $C_{j'}$ lie in the same component in $\widetilde{F}_\varepsilon$.

Recall that after applying Lemma 9 to the input, the ratio between the shortest and longest edge is at most $2n/\varepsilon$, where n is the number of vertices of the original input graph. Since we assume that the length of the shortest edge is 1, the cost of any component lies between 1 and $2n^2/\varepsilon$, given that a component is a tree with less than n edges. Therefore $\lfloor\operatorname{cost}(C_j)\rfloor_2 \in \{2^q \mid q \in \mathbb{N}_0 \wedge 0 \leq q \leq \log_2(2n^2/\varepsilon)\}$, which means that the algorithm will consider the above sequence in some iteration.

We now turn to $\pi = \{Y_1,\dots,Y_{|\rho|}\}$ constructed for this sequence, and show that it is a partition of A_B and that $\widetilde{F}_\varepsilon$ conforms to it. For this, note that no set of ζ_B contains net points of several active components of F_ε, since by Lemma 10 all active terminals in the same set of ζ_B also belong to the same component of F_ε. Thus the sets S_j as defined above (and also the corresponding sets U_j) are pairwise disjoint. This means that, due to the definition of ρ, any two terminals $t \in U_j$ and $t' \in U_{j'}$ end up in the same set of π if and only if t and t' belong to the same component of $\widetilde{F}_\varepsilon$ (as $U_j \subseteq Y_{\rho(j)}$).

Now consider a terminal $t \in A_B$, which does not lie in any U_j, and let q be the index of the active component C_q of F_ε containing t. As $\delta_q = \lfloor\operatorname{cost}(C_q)\rfloor_2$, N_q is an $\frac{\varepsilon}{k}\delta_q$-net of $C_q \cap A_B$. Also, we chose S_q so that $N_q \subseteq U_q$. Hence we get $\operatorname{dist}(t,U_q) \leq \operatorname{dist}(t,N_q) \leq \frac{\varepsilon}{k}\delta_q$, and the definition of p_t implies $p_t \leq q$. Now C_{p_t} is either equal to C_q, or C_q is connected to the component C_{p_t} in the approximate solution $\widetilde{F}_\varepsilon$ according to Rule 2: on one hand we have $C_{p_t} \leq C_q$ due to the order of the indices, and at the same time by Lemma 10 we have $U_{p_t} \subseteq V(C_{p_t}) \cap A_B$, which implies

$$\operatorname{dist}(C_q,C_{p_t}) \leq \operatorname{dist}(t,C_{p_t}) \leq \operatorname{dist}(t,U_{p_t}) \leq \frac{\varepsilon}{k}\delta_{p_t} = \frac{\varepsilon}{k}\lfloor\operatorname{cost}(C_{p_t})\rfloor_2.$$

Hence we can conclude that t lies in the same component as C_{p_t} in $\widetilde{F}_\varepsilon$.

In conclusion, adding U_j to $Y_{\rho(j)}$ and t to $Y_{\rho(p_t)}$ for each terminal t not lying in any U_j, partitions the terminals according to the components of $\widetilde{F}_\varepsilon$. Hence π is a partition of the active terminals A_B that is added to Π_B, and $\widetilde{F}_\varepsilon$ conforms to it. □

Using all of the above, we can finally prove our main theorem, stating that there is an EPAS for Steiner Forest parameterized by the treewidth.

Proof of Theorem 1. The first steps of our algorithm are to preprocess the given tree decomposition using Lemma 8 so that it is nice and its height is $O(k \log n)$, and the input graph using Lemma 9 so that the aspect ratio is bounded (which means that n denotes the number of vertices in the original input graph). We then compute the partition sets Π_B for all bags B using the above procedure, resulting in partition sets of size $(\frac{k}{\varepsilon} \log \frac{n}{\varepsilon})^{O(k^2/\varepsilon)} = 2^{O(\frac{k^2}{\varepsilon} \log \frac{k}{\varepsilon})} \cdot n^{o(1)}$. Here, we are using a well-known Win/Win argument: if $k^2/\varepsilon < \sqrt{\log n}$, then $(\log n)^{k^2/\varepsilon} = n^{o(1)}$; otherwise, $\log n \leq k^4/\varepsilon^2$, therefore $(\frac{k}{\varepsilon} \log \frac{n}{\varepsilon})^{O(k^2/\varepsilon)} = (\frac{k}{\varepsilon})^{O(\frac{k^2}{\varepsilon})}$.

Since each partition of a set Π_B can be computed in polynomial time, and the number of bags of the nice tree decomposition is $O(kn)$, this takes $2^{O(\frac{k^2}{\varepsilon} \log \frac{k}{\varepsilon})} \cdot n^{O(1)}$ time. Next we apply Theorem 5 to compute a solution that is at least as good as $\widetilde{F}_\varepsilon$ conforming to all Π_B, in $2^{O(\frac{k^2}{\varepsilon} \log \frac{k}{\varepsilon})} \cdot n^{O(1)}$ time. Hence we obtain a $(1+\varepsilon)^2$-approximation F. According to Lemma 9, F can be converted into a $((1+\varepsilon)^2+\varepsilon)$-approximation to the original input graph. Since for any $\varepsilon' > 0$ we may choose $\varepsilon = \Theta(\varepsilon')$ so that $((1+\varepsilon)^2+\varepsilon) \leq 1+\varepsilon'$, we obtain an EPAS as claimed. □

4 Vertex cover

In this section we consider the parameterization by the size of a *vertex cover*, which is a set $S \subseteq V$ of vertices such that every edge is incident on at least one of the vertices of S. We first present an easy FPT algorithm based on the dynamic program given by Theorem 5, and then prove that its runtime dependence on the parameter is asymptotically optimal.

4.1 FPT Algorithm

Let $S \subseteq V$ be a given vertex cover of size k for the input graph G. We will show how to construct a tree decomposition with all required properties of Theorem 5 in order to run the corresponding dynamic program. For this the tree decomposition $(T, \{B_i\}_{i \in V(T)})$ needs to be *nice*.

We may assume w.l.o.g. that the vertex cover S contains no terminal: using a standard preprocessing procedure, we can replace any terminal $t \in S$ of the vertex cover by a Steiner vertex v and then connect t with v using an edge of cost 0. Note that S is still a vertex cover for the preprocessed graph and that the complement set $I = V \setminus S$ of the vertex cover is an independent set containing all terminals. To use Theorem 5, we will first construct a (trivial) nice tree decomposition for I and then add S to each bag.

Note that a terminal $t \in R$ can be part of several demand pairs of the Steiner Forest instance. Consider the *demand graph* H with vertex set R and an edge for each demand pair. Any subset of R that induces a maximal connected component of H is called a *group*. Note that every group of terminals must lie in the same connected component of any Steiner Forest solution. For each group $R' \subseteq R$, we create one leaf node of the tree decomposition for each terminal $t \in R'$ and let the corresponding bag contain t. We then add a forget node for each such leaf node, which we add as parent to the leaf with an empty bag. These forget nodes are then connected in a binary tree by adding join nodes with empty bags (unless the group only contains one terminal in which case we skip this step). We proceed in the same way for the Steiner vertices of the independent set I, that is, if we consider $I \setminus R$ to be a group as well we obtain a nice tree decomposition for $I \setminus R$ in which each bag of a leaf node contains one vertex of $I \setminus R$. All these trees are then connected using join nodes with empty bags, to obtain a tree decomposition (of width 0) for the independent set I. Finally, we simply add the vertex cover S to every bag, which results in a nice tree decomposition (of width k) for the graph G, such that every terminal lies in a bag of a leaf node (as $S \cap R = \emptyset$).

In the obtained tree decomposition, let V_B be the vertices of G contained in all bags in the subtree rooted at the node associated with B. By construction, V_B either contains no terminals (if B is a bag of the tree decomposition for $I \setminus R$), fully contains some groups of R (if B is the bag of the root of a tree decomposition for a group $R' \subseteq R$, or if B is a bag of a join node used to connect the tree decompositions for groups in the last step), or contains some strict subset of only one group of R (if B is a bag of a

non-root node of a tree decomposition for a group $R' \subseteq R$). If no terminals lie in V_B then clearly there are no active terminals for bag B. However this is also the case if V_B fully contains some groups of R. Hence in both these cases the set Π_B of permutations of active terminals is empty. Whenever V_B contains a strict subset of only one group $R' \subseteq R$, the active terminals A_B of B are only from this set, i.e., $A_B \subseteq R'$. Thus we can add the trivial partition $\pi = \{A_B\}$ as the only partition of Π_B, since all terminals of R' belong to the same component of any solution, including the optimum.

Clearly, the optimal solution conforms with these sets Π_B of permutations, and the total number p of permutations is at most the number of groups, which is at most $n/2$. Hence by Theorem 5 we obtain the algorithm of Theorem 4.

4.2 Runtime lower bound

Our goal here is to present a reduction showing that the algorithm we have given for Steiner Forest parameterized by vertex cover is essentially optimal, assuming the ETH. Recall that the ETH is the hypothesis that 3-SAT on instances with n variables cannot be solved in time $2^{o(n)}$. We will give a reduction that given a 3-SAT instance ϕ, produces an equivalent Steiner Forest instance with vertex cover at most $O(n/\log n)$. We stress that our reduction works even for unweighted instances.

Theorem 17. *If there exists an algorithm which, given an unweighted Steiner Forest instance on n vertices with vertex cover k, finds an optimal solution in time $2^{o(k \log k)} n^{O(1)}$, then the ETH is false.*

Proof. We present a reduction from 3-SAT. Before we proceed, we would like to add to our formula the requirement that the variable set comes partitioned into three sets in a way that each clause contains at most one variable from each set. It is not hard to show that this does not affect the complexity of the instance much, as we demonstrate in the following claim.

Claim 18. *Suppose that there exists an algorithm that takes as input a 3-SAT instance ϕ on n variables and a partition of the variables into three sets of equal size, such that each clause contains at most one variable from each set, and decides if ϕ is satisfiable in time $2^{o(n)}$. Then, the ETH is false.*

Proof. Suppose we start with an arbitrary 3-SAT formula ψ on n variables $x_1, \ldots, x_n$. Under the ETH, it should be impossible to decide if ψ is satisfiable in time $2^{o(n)}$. We will edit ψ to produce the partition of the variables into three sets. For each variable x_i, we introduce two new variables x'_i, x''_i, and add to the formula the clauses $(x_i \to x'_i) \wedge (x'_i \to x''_i) \wedge (x''_i \to x_i)$. The variables of ψ can now be partitioned into three sets $X = \{x_1, \ldots, x_n\}$, $X' = \{x'_1, \ldots, x'_n\}$, and $X'' = \{x''_1, \ldots, x''_n\}$. Furthermore, because of the clauses we added it is not hard to see that in any satisfying assignment x_i, x'_i, and x''_i must be given the same value. We then repeat the following: as long as there exists a clause that contains more than one variable from X, arbitrarily pick a literal of this clause that contains $x_i \in X$ and replace in it x_i by x'_i or x''_i, in a way that the clause contains at most one variable from each group. The new formula we have constructed in this way is equisatisfiable to ψ, has $n' = O(n)$ variables and $O(n+m)$ clauses, and its variables are partitioned into three sets so that each clause contains at most one variable from each set. Therefore, the new formula cannot be solved in time $2^{o(n')}$ under the ETH. □

In the remainder we will then assume that we are given a formula ϕ on $3n$ variables which are partitioned into three sets of size n as specified by the previous claim. Without loss of generality, suppose that n is a power of 4 (this can be achieved by adding dummy variables). Note that this ensures that $\frac{\log n}{2}$ and $\sqrt{n}$ are both integers.

We construct an equivalent instance of Steiner Forest as follows. Let $L = \lceil \frac{n}{\log^2 n} \rceil$. We begin by constructing i choice gadgets, i.e., for $i \in \{1, \ldots, 3 \log n\}$ we make:

- $2L$ left vertices, labeled ℓ^i_j, for $j \in \{0, \ldots, 2L-1\}$.
- $2L$ right vertices, labeled r^i_j, for $j \in \{0, \ldots, 2L-1\}$.
- $\sqrt{n}$ middle vertices, labeled m^i_j, for $j \in \{0, \ldots, \sqrt{n}-1\}$.
- We connect all middle vertices to all left and right vertices, that is, for all $j \in \{0, \ldots, 2L-1\}$ and $j' \in \{0, \sqrt{n}-1\}$ we connect ℓ^i_j and r^i_j to $m^i_{j'}$.
- For each $j \in \{0, \ldots, 2L-1\}$ we add a demand from ℓ^i_j to r^i_j.

Notice that the graph we have constructed so far contains $3\log n$ choice gadgets, each of which has $4L + \sqrt{n} = O(n/\log^2 n)$ vertices, so the graph at the moment contains $O(n/\log n)$ vertices in total.

Before we proceed, let $X = X_a \cup X_b \cup X_c$ be the set of $3n$ variables of ϕ that was given to us partitioned into three sets of size n. We partition X into $3\log n$ groups $X_1, \ldots, X_{3\log n}$ in a way that (i) $|X_i| \leq \lceil n/\log n \rceil$ for all $i \in \{1, \ldots, \log n\}$ and (ii) for all $i \in \{1, \ldots, \log n\}$ we have X_i is contained in one of X_a, X_b, X_c. This can be done by taking the n variables of X_a and partitioning them arbitrarily into groups $X_1, \ldots, X_{\log n}$ of size as equal as possible (therefore at most $\lceil n/\log n \rceil$), and we proceed similarly for X_b, X_c. Rename the variables of ϕ so that for each i we have that $X_i = \{x_{(i,0)}, \ldots, x_{(i,\lceil n/\log n \rceil - 1)}\}$.

To give some intuition, we will now say that, for $i \in \{1, \ldots, 3\log n\}$, the choice gadget i represents the variables of the set X_i. In particular, for each $j \in \{0, \ldots 2L-1\}$, we will say that the way that the demand $\ell^i_j \to r^i_j$ was satisfied encodes the assignment to the $\frac{\log n}{2}$ variables $\{x_{(i,\frac{j\log n}{2})}, \ldots, x_{(i,\frac{(j+1)\log n}{2}-1)}\}$. More precisely, in our intended solution the demand $\ell^i_j \to r^i_j$ is satisfied by connecting both terminals to a common middle vertex $m^i_{j'}$. We can infer the assignment to the $\frac{\log n}{2}$ variables this represents simply by writing down the binary representation of j', which is a number between 0 and $\sqrt{n}-1$, hence a number with $\frac{\log n}{2}$ bits. Note that this way we represent $2L \cdot \frac{\log n}{2} \geq \lceil \frac{n}{\log n} \rceil$ variables, that is, we can represent the assignment to all the variables of the group.

Armed with this intuition, we can now complete our construction. For each clause c we construct two new vertices, c_1, c_2 and add a demand from c_1 to c_2. For each literal contained in c, suppose that the literal involves the variable $x_{(i,\frac{j\log n}{2}+\alpha)}$ for $i \in \{1, \ldots, 3\log n\}$, $j \in \{0, \ldots, 2L-1\}$, $\alpha \in \{0, \ldots, \frac{\log n}{2}-1\}$. We then connect c_1 to ℓ^i_j. Furthermore, if $x_{(i,\frac{j\log n}{2}+\alpha)}$ appears positive in c, we connect c_2 to all $m^i_{j'}$ such that the binary representation of j' has a 1 in position α. If on the other hand $x_{(i,\frac{j\log n}{2}+\alpha)}$ appears negative in c, we connect c_2 to all $m^i_{j'}$ such that the binary representation of j' has a 0 in position α. In other words, we connect c_2 to all the middle vertices to which ℓ^i_j could be connected and are consistent with an assignment that satisfies c using the current literal. After repeating the above for all literals of each clause the construction is complete. We set the target cost to be $B = 2m + 12L\log n$.

Before we argue about the correctness of the reduction, let us observe that if the reduction preserves the satisfiability of ϕ, then we obtain the theorem, because the instance we constructed has vertex cover $k = O(n/\log n)$ and size polynomial in the size of ϕ. Indeed, as we argued the choice gadgets have $O(n/\log n)$ vertices in total, and all further edges we added have an endpoint in a choice gadget. If there was an algorithm solving the new instance in time $k^{o(k)}n^{O(1)}$, this would give a $2^{o(n)}$ algorithm to decide ϕ.

Regarding correctness, let us first observe that if ϕ is satisfiable, we can obtain a valid solution using the intuitive translation from assignments to choice gadget solutions we gave above. In particular, for each $i \in \{1, \ldots, 3\log n\}$ and $j \in \{0, \ldots, 2L-1\}$, we consider the assignment to variables $\{x_{(i,\frac{j\log n}{2})}, \ldots, x_{(i,\frac{(j+1)\log n}{2}-1)}\}$ as a binary number, which must have a value j' between 0 and $\sqrt{n}-1$. We then connect both ℓ^i_j, r^i_j to $m^i_{j'}$. Repeating this satisfies all demands internal to choice gadgets and uses $3\log n \cdot 4L = 12L\log n$ edges. Consider now a clause c and the demand from c_1 to c_2. Since we started with a satisfying assignment, c must contain a true literal, say involving the variable $x_{(i,\frac{j\log n}{2}+\alpha)}$. We select the edge from c_1 to ℓ^i_j. Furthermore, we observe that c_2 must be a neighbor of all vertices $m^i_{j'}$ such that the bit in position α of the binary representation of j' agrees with the value of $x_{(i,\frac{j\log n}{2}+\alpha)}$. Since ℓ^i_j is already connected to such a $m^i_{j'}$, we select the edge from that vertex to c_2 to satisfy the demand for this clause. We have therefore spent $2m$ further edges for the clause demands and have used a budget of exactly B.

For the converse direction, suppose we have a solution of cost B. We first observe that each vertex r^i_j must be connected to a middle vertex $m^i_{j'}$, since all right vertices are terminals, but such vertices only have edges connecting them to middle vertices. Recall that, for each i, j, the left vertex ℓ^i_j must be in the same component of the solution as r^i_j, since there is a demand between these two vertices. Hence, each ℓ^i_j is in the same component of the solution as some $m^i_{j'}$. We now slightly edit the solution as follows: suppose there exists a vertex ℓ^i_j which is not directly connected in the solution to any middle vertex $m^i_{j'}$. Since this vertex is in the same component as one such vertex $m^i_{j'}$, we add to the solution the edge connecting them, and since this creates a cycle, remove from the solution another edge incident on ℓ^i_j. Doing this repeatedly ensures that each ℓ^i_j is connected to a middle vertex $m^i_{j'}$ in the solution without increasing the total cost.

We now observe that since each ℓ^i_j and each r^i_j is connected to at least one middle vertex $m^i_{j'}$ in the solution, this already uses a cost of $3\log n \cdot 4L = 12L\log n$. Furthermore, for each clause we have

constructed two terminals, each of which must use at least one of its incident edges, giving an extra cost of $2m$. Since our budget is exactly $2m + 12L \log n$, we conclude that each terminal constructed for a clause is incident on exactly one edge, and each ℓ^i_j and each r^i_j is connected to exactly one middle vertex. Crucially, these observations imply the following fact: if for some i, j, j' we have that ℓ^i_j and $m^i_{j'}$ are in the same component of the solution, then the edge connecting ℓ^i_j and $m^i_{j'}$ is part of the solution. To see this, observe that any path connecting ℓ^i_j and $m^i_{j'}$ that is not a direct edge would need to have length at least 3. However, no clause terminal can be an internal vertex of such a path, since clause terminals have degree 1 in the solution. Furthermore, if we remove clause terminals from the graph, left and right vertices also have degree 1 in the remaining solution, so such vertices also cannot be internal in the path. Finally, middle vertices are an independent set, so it is impossible for all internal vertices of a path of length at least 3 to be middle vertices.

Armed with the observation that ℓ^i_j and $m^i_{j'}$ are in the same connected component of the solution if and only if they are directly connected, we are ready to extract a satisfying assignment from the Steiner forest. For each i, j, if ℓ^i_j is connected to $m^i_{j'}$ we write j' in binary and assign to variable $x_{(i, \frac{j \log n}{2} + \alpha)}$, for $\alpha \in \{0, \ldots, \frac{logn}{2} - 1\}$ the value in position α of the binary representation of j'. We claim that this assignment must be satisfying. Indeed, consider the clause c, and the terminals c_1, c_2 which represent it. Since these terminals have a demand, they must be in the same component. Because c_1 has at most three neighbors which are in different choice gadgets (as each clause contains variables from distinct groups), we can see that c_1 must be connected to some ℓ^i_j and c_2 to some $m^i_{j'}$ in the solution, such that ℓ^i_j and $m^i_{j'}$ are in the same component, and are therefore directly connected. But if ℓ^i_j is directly connected to $m^i_{j'}$ this means that the assignment we extracted from ℓ^i_j gives a value to a variable $x_{(i, \frac{j \log n}{2} + \alpha)}$ which satisfies the clause c, hence we have a satisfying assignment. □

5 Feedback Edge Set

A *feedback edge set* of a graph is a set of edges that when removed renders the graph acyclic. It is well-known that if G is a connected undirected graph on n vertices and m edges, then all minimal feedback edge sets of G have size $k = m - n + 1$. Indeed, such a set can be constructed in polynomial time by repeatedly locating a cycle in the graph and selecting an arbitrary edge of the cycle to insert into the feedback edge set.

In this section we will consider Steiner Forest parameterized by the feedback edge set of the input graph, which we will denote by k. Unlike the vertex cover section, here our main result is positive: we show that Steiner Forest can be solved optimally in time $2^{O(k)}n^{O(1)}$, that is, in time single-exponential in the parameter. Since we are able to achieve a single-exponential dependence, it is straightforward to see that this is optimal under the ETH.

Theorem 19. *If there is an algorithm solving Steiner Tree in time $2^{o(k)}n^{O(1)}$, where k is the feedback edge set of the input, then the ETH is false.*

Proof. The proof follows from the sparsification lemma of Impagliazzo, Paturi, and Zane [25] composed with the standard reduction proving that Steiner Tree is NP-complete. We sketch the details. Suppose we are given a 3-SAT formula ϕ with n variables and m clauses. The sparsification lemma shows that in order to disprove the ETH it is sufficient to show that we can decide if ϕ is satisfiable in time $2^{o(n)}$ under the restriction that $m = \Theta(n)$. We edit ϕ to obtain an equisatisfiable formula ϕ' where every variable appears at most 3 times (for each variable x appearing $f > 3$ times, we replace each occurrence of x with f fresh variables $x_1, \ldots, x_f$ and add the clauses $(x_1 \to x_2) \wedge (x_2 \to x_3) \wedge \ldots (x_f \to x_1)$). The new formula ϕ' has $n' = O(n)$ variables and $m' = O(n)$ clauses. We now execute the chain of reductions showing that Steiner Tree is NP-hard (e.g. from [26]), which produce an instance on a graph $G = (V, E)$ with $|E| = O(m')$, therefore, $|E| = O(n)$. The new instance has feedback edge set size $k < |E|$, therefore an algorithm solving the new instance in time $2^{o(k)}|V|^{O(1)}$ would falsify the ETH. □

Let us now proceed to the detailed presentation of the algorithm. Suppose that we are given a budget b and we want to decide if there exists a Steiner Forest solution F such that $\text{cost}(F) \leq b$. We start by applying a simple reduction rule.

Rule 3: Suppose we have a Steiner Forest instance on graph G with weight function w and budget b, such that a vertex $u \in V$ has degree 1. If $u \notin R$, then delete u. If $u \in R$, let v be the unique neighbor of u. Then set $b' := b - w(uv)$, delete u from the graph and the demand $\{u, v\}$ from D

if it exists, and replace, for each $x \in V \setminus \{u, v\}$ such that $\{u, x\} \in D$ the demand $\{u, x\}$ with the demand $\{v, x\}$.

Lemma 20. *Rule 3 is safe.*

Proof. If $u \notin R$ then no optimal solution contains the edge uv, so it is safe to delete u. If $u \in R$ then all feasible solutions contain the edge uv. □

Observe that if we apply Rule 3 exhaustively, then the minimum degree of the graph is 2. As we show next, relatively few vertices can have higher degree.

Lemma 21. *Suppose we have a* Steiner Forest *instance with feedback edge set of size k and minimum degree at least 2. Then G contains at most $2k$ vertices of degree at least 3.*

Proof. We observe that if our graph has a feedback edge set of size k, then $m = k + n - c$, where c is the number of connected components of G. This implies that $\sum_{v \in V} d(v) = 2m = 2k + 2n - 2c$. Let V_2 be the set of vertices of degree exactly 2 and $V_3 = V \setminus V_2$ be the set of vertices of degree at least 3. We have $\sum_{v \in V} d(v) \geq 2|V_2| + 3|V_3| = 2n + |V_3|$. We conclude that $|V_3| \leq 2k - 2c$. □

In the remainder we will assume that we have a Steiner Forest instance $G = (V, E)$ with a feedback edge set $H \subseteq E$ of size k, to which Rule 3 can no longer be applied. We will say that a vertex v is *special* if v is incident on an edge of H or v has degree at least 3. By Lemma 21 we know that G contains at most $4k$ special vertices.

We define a *topological edge* (topo-edge for short) as follows: a path P in G is a topological edge if the two endpoints of P are special vertices and all internal vertices of P are non-special. Note that by this definition, all edges of H form topo-edges, since the endpoints of such edges are special. We observe the following:

Lemma 22. *Suppose we have a graph G with feedback edge set of size k and minimum degree at least 2. Then G contains at most $5k$ topological edges.*

Proof. Let V_s be the set of special vertices and $V_t = V \setminus V_s$. If we have more than $5k$ topological edges in G then $\sum_{v \in V_s} d(v) \geq 10k$. This is because each topological edge contributes at least 2 in the sum $\sum_{v \in V_s} d(v)$. On the other hand, if c is the number of connected components of G, we have $2n + 2k - 2c = 2m = \sum_{v \in V} d(v) = \sum_{v \in V_s} d(v) + \sum_{v \in V_t} d(v) = \sum_{v \in V_s} d(v) + 2|V_t|$. However, $|V_t| \geq n - 4k$ by Lemma 21 and the fact that at most $2k$ vertices are incident on H. Hence, $2n + 2k - 2c \geq \sum_{v \in V_s} d(v) + 2n - 8k$. This implies that $\sum_{v \in V_s} d(v) < 10k$. Hence, it is impossible to have more than $5k$ topological edges. □

We are now ready to state the main algorithmic result of this section.

Theorem 23. *There is an algorithm that solves* Steiner Forest *on instances with n vertices and a feedback edge set of size k in $2^{O(k)} n^{O(1)}$ time.*

Proof. Call the set of special vertices V_s and let $V_t = V \setminus V_s$. For the rest of this proof and for the sake of the analysis, fix an optimal solution $F^\star$.

To begin, we guess which of the $5k$ topological edges according to Lemma 22 are fully used in the optimal solution. To be more precise, we will say that a topo-edge P is fully used in $F^\star$ if all edges of the path P are contained in $F^\star$. This gives 2^{5k} possibilities. In the remainder we will assume that we have correctly guessed the set of topo-edges which are fully used in $F^\star$.

We now observe that for any two vertices $u, v \in V_s$ we have enough information to deduce whether u, v are in the same connected component of $F^\star$. More precisely, we construct an auxiliary graph G_s with vertex set V_s that contains an edge between two vertices $u, v \in V_s$ if there exists a fully used topo-edge whose endpoints are u, v. We now claim that two vertices $u, v \in V_s$ are in the same component of $F^\star$ if and only if u, v are in the same connected component of G_s. Indeed, if two vertices u, v are in the same component of G_s, then clearly there is a path connecting them in $F^\star$ going through fully used topo-edges; conversely if u, v are in the same component of $F^\star$ and the path connecting them goes through the special vertices $u_1 = u, u_2, \ldots, u_\ell = v$ (and all other vertices are non-special), then the path $u_1, \ldots, u_\ell$ also exists in G_s, as the topo-edge connecting u_i, u_{i+1} must be fully used.

Because of the above, we can now assume that we have a partition ρ of V_s such that u, v are in the same set of ρ if and only if u, v are in the same connected component of $F^\star$. Notice that this implies that we can remove from the instance all demands $\{u, v\} \in D$ such that $u, v \in V_s$: if u, v are in the same set of ρ the demand is automatically satisfied by our guess of the fully used topo-edges; while if u, v

are in distinct sets of ρ, we know that our guess is incorrect and we reject the current instance. Every remaining demand of our instance is therefore incident on at least one non-special vertex.

What remains is to decide for topo-edges which are not fully used, which of their incident edges belong in $F^\star$. Note that this is trivial for topo-edges consisting only of a single edge, since fully using such a topo-edge is equivalent to placing the corresponding edge in the solution. We therefore focus on topo-edges which contain at least one internal (non-special) vertex.

For this we proceed in several steps. First, suppose we have a non-fully-used topo-edge P whose endpoints are adjacent to $u, v \in V_s$ such that u, v are in the same component of $F^\star$. We edit the instance so that demands with one endpoint in the interior of P also have their other endpoint in P. More precisely, for each demand $\{x, y\} \in D$ such that x is an internal vertex of P and $y \notin P$, we remove $\{x, y\}$ from D and replace it with the demands $\{x, u\}$ and $\{y, u\}$. It is not hard to see that this is safe, because any path satisfying the demand $\{x, y\}$ would have to go either through u or through v, but u, v are in the same component of $F^\star$ thanks to other, fully-used topo-edges, so routing the demand through u or v is the same.

Consider then a topo-edge P whose endpoints u, v are in the same component of $F^\star$ and where all demands with one endpoint in an internal vertex of P have the other endpoint in P. We simplify the instance by branching: select an edge $e \in P$, delete e from the instance, and apply Rule 3 exhaustively on internal vertices of P, until all such vertices are removed. Since we have guessed that P is not fully used, at least one of the instances we produced is equivalent to the original, that is, at least one choice of edge to delete indeed deletes an edge not used by the optimal solution. It may seem that since we are branching on n possibilities, this branching will lead to a running time of n^k. However, we observe that after removing any edge of P and exhaustively applying Rule 3 we obtain instances which have (i) the same graph, as all internal vertices of P have been deleted and all other vertices are unchanged (ii) the same set of demands, as all demands with one endpoint in an internal vertex of P have either been removed or replaced with the demand $\{u, v\}$ (which is satisfied by the fully used topo-edges, so can be removed) and other demands are unchanged. Hence, among the at most n instances this branching produces, it suffices to select the one with the maximum remaining budget and solve that, to decide if the original is a Yes instance. In other words, the branching procedure of this paragraph is a polynomial-time reduction rule which allows us to eliminate all topo-edges whose endpoints are in the same component of $F^\star$.

In the remainder we thus assume that every topo-edge P that is not fully used has endpoints $u, v \in V_s$ which are in distinct components of $F^\star$. Next, we deal with the case of "internal" demands. Suppose that there exists a topo-edge P with endpoints $u, v \in V_s$ that contains an internal demand, that is, there exist $w_1, w_2 \in P \setminus \{u, v\}$ such that $\{w_1, w_2\} \in D$. Then, all edges in the path from w_1 to w_2 in P must belong in $F^\star$, because every other solution that connects w_1 to w_2 would put u, v in the same component. We can therefore contract all the edges of the path from w_1 to w_2 and adjust our budget and our demands accordingly: we decrease our budget by the total cost of the edges of the path from w_1 to w_2, we remove all demands that have both endpoints in that path, and for demands that have one endpoint in that path, we replace that endpoint by the vertex that results from the contraction of the path.

We now arrive at the case where the endpoints of each topo-edge are adjacent to vertices from distinct components of $F^\star$ and demands with one endpoint in the interior of a topo-edge have the other endpoint outside of the topo-edge or in V_s.

We distinguish several cases:

1. There exists a topo-edge P adjacent to $u, v \in V_s$, an internal vertex $w \in P \setminus \{u, v\}$ and a vertex $w' \in V_s$ such that $\{w, w'\} \in D$. If w' is in the same component of $F^\star$ as u (respectively v), we include in the solution all edges in the path in P from w to u (respectively v), contract the selected edges and update our budget and demands accordingly, as above. If w', u, v are in distinct components of $F^\star$, then we conclude that the current guess is incorrect and reject the instance. Correctness of these actions follows if we assume that the partition ρ of V_s we have computed corresponds to the connected components of $F^\star$, because in the latter case any solution that connects w to w' will place w' in the same component as one of u, v, and in the former case we are forced to use the selected path, as otherwise u, v would end up in the same component of $F^\star$.

2. There exist two topo-edges P_1, P_2 adjacent to $u_1, v_1 \in V_s$ and $u_2, v_2 \in V_s$ respectively, and vertices $w_1 \in P_1$ and $w_2 \in P_2$ such that $\{w_1, w_2\} \in D$. If u_1, v_1, u_2, v_2 are in four distinct components of $F^\star$ we reject the current guess, as it is impossible to place w_1, w_2 in the same component without also placing some of u_1, v_1, u_2, v_2 in the same component.

3. If $u_1, u_2, v_1, v_2, w_1, w_2$ are as previously but u_1, v_1, u_2, v_2 are in three distinct components of $F^\star$, we can assume without loss of generality that u_1, u_2 are in the same component. We replace the demand $\{w_1, w_2\}$ with the demands $\{w_1, u_1\}$ and $\{w_2, u_2\}$ and reduce to a previous case.

Finally, if none of the previous cases apply we have arrived at an instance where all remaining demands $\{w_1, w_2\} \in D$ satisfy that w_1, w_2 belong in two distinct topo-edges P_1, P_2, which are incident on $u_1, v_1 \in V_s$ and $u_2, v_2 \in V_s$ respectively, such that u_1, u_2 are in the same component of $F^\star$, and so are v_1, v_2, but the component of u_1, u_2 is distinct from the component of v_1, v_2. We will find the best way to satisfy such demands by solving an auxiliary problem.

Fix two sets C_1, C_2 of the partition ρ of V_s which we have computed and consider every topo-edge P with one endpoint in C_1 and the other in C_2. We construct a new instance of Steiner Forest on a graph G_2 by taking the union of all such topo-edges and then contracting all vertices of C_1 into a single vertex c_1 and all vertices of C_2 into a single vertex c_2. We include in the new instance all demands with at least one endpoint on one of the internal vertices of the topo-edges we used; note that such demands also have the second endpoint in G_2. Let G_1 be the instance induced from the original graph if we delete all internal topo-edge vertices which appear in G_2. Note that every demand of the original instance appears in either G_1 or G_2.

We will now state two claims:

Claim 24. *If the optimal Steiner Forest solution on the instance G_2 constructed above has cost b_2, then we have the following: G has a solution of cost at most b consistent with the guess ρ if and only if G_1 has a solution consistent with the guess ρ of cost at most $b - b_2$.*

Claim 25. *The optimal solution to G_2 can be computed in polynomial time by a reduction to the Min Cut problem.*

Let us explain why the claims are sufficient to conclude our algorithm. We consider every pair of sets $C_1, C_2 \in \rho$ (of which there are $O(k^2)$) and for each such pair the claims imply that we can decompose the instance into two instances G_1, G_2, such that G_2 can be solved in polynomial time, and using the optimal value we calculate for G_2 we can reduce solving G to solving G_1. Repeating this for all pairs results in an instance with no demands. Putting everything together, we have that for one of 2^{5k} possible guesses (on which topo-edges are fully used) we apply a series of polynomial-time reduction rules that allow us to decompose the instance into $O(k^2)$ polynomial-time solvable sub-problems. We therefore obtain an exact algorithm running in $2^{O(k)}n^{O(1)}$ time. □

Proof of Claim 24. If G_1 has a solution of cost $b - b_2$ consistent with ρ, then we can form a solution for G by taking the union of the solution for G_1 with an optimal solution for G_2. This will have cost at most b. Furthermore, recall that all demands of G appear in either G_1 or G_2. Demands that appear in G_1 are clearly satisfied by the new solution in G, while demands that appear in G_2 are satisfied because the solution in G_1 is consistent with ρ, so it contains paths between any two $u, v \in C_1$ for each $C_1 \in \rho$.

For the converse direction, suppose G has a solution of cost b consistent with ρ. We observe that this solution restricted to G_2 is a feasible solution (which furthermore places c_1, c_2 in distinct components), hence must have cost at least b_2. Therefore, the solution restricted to edges of G_1 has cost at most $b - b_2$. Because all topo-edges included in G_2 are topo-edges which are not fully used (according to the guess that gave us ρ), the solution we construct in G_1 is still consistent with ρ and satisfies all demands. □

Proof of Claim 25. Before we begin, we perform a basic simplification step. If the instance contains a Steiner vertex v of degree 2 (that is, a vertex not incident on any demand), with neighbors u_1, u_2, then we remove v from the instance and add an edge u_1u_2 with weight equal to $w(vu_1) + w(vu_2)$. It is not hard to see that the new instance is equivalent (v would only be used in a solution if both its incident edges are used), and we now know that all vertices of degree 2 are terminals.

Recall that we have a graph G_2 with two special vertices c_1, c_2 such that the graph consists of a collection of parallel paths with endpoints c_1, c_2, and furthermore every demand is between two internal vertices of distinct paths. For the purposes of the larger algorithm we are interested in computing the best solution where c_1, c_2 are in distinct components, but for the sake of completeness let us briefly note that G_2 can be solved to optimality without this constraint, as the best solution where c_1, c_2 are in the same component is just a minimum cost spanning tree of G_2 (here we are using the fact that all internal vertices of all paths are terminals).

In order to compute the best solution that places c_1, c_2 into distinct components, we will reduce the problem to Min Cut. Let N be a sufficiently large value, for example set N to be the sum of all edge weights of the instance. We construct a Min Cut instance on the same graph but with weight function

$w'(e) = N - w(e)$. Furthermore, for all w_1, w_2 such that $\{w_1, w_2\} \in D$ we add an edge w_1w_2 and set $w'(w_1w_2) = n^2N$.

Our claim is now that if F_c is a set of edges that gives a minimum weight $c_1 - c_2$ cut in the new instance, then the complement of F_c is a minimum cost Steiner Forest solution for G_2 that places c_1, c_2 in distinct components.

To prove the claim, suppose that F_c is a minimum-weight $c_1 - c_2$ cut in the new graph. We observe that F_c cannot include any of the edges we added between the endpoints of demands $(w_1, w_2) \in D$, as such edges have a very high cost (deleting every other edge would be cheaper). Furthermore, because all edges have positive weight and the cut F_c is minimal, removing F_c from the graph must leave exactly two connected components, one containing each of c_1, c_2. Hence, for each $(w_1, w_2) \in D$, if we keep in the graph all edges not in F_c, w_1, w_2 are in the same component, and we have a feasible Steiner Forest solution for all the demands. In the other direction, consider an optimal Steiner Forest solution that places c_1, c_2 in distinct components, and let F'_c be the set of edges of G_2 *not* included in the solution. F'_c must be a valid $c_1 - c_2$ cut, because it contains at least one edge from each topological edge connecting c_1 to c_2 (otherwise c_1, c_2 would be in the same component); and as each demand $(w_1, w_2) \in D$ is satisfied, therefore, w_1, w_2 are either in the component of c_1 or in the component of c_2. We have therefore established a one-to-one mapping between optimal minimum cuts and optimal Steiner Forest solutions and conclude the claim by observing that by minimizing the weight of F_c in the Min Cut instance, we are maximizing the weight of non-selected edges in the Steiner Forest instance (thanks to the modified weight function), hence we are selecting an optimal Steiner Forest solution. □

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